Ice breaker:

Why Linear Algebra?

What is the most important thing you have learned this semester?

What is a matrix?

What are matrices used for?

What is the meaning of $A\vec{x} = \vec{b}$?

What do we use this for?

What is orthogonality?
Why is it important?

What is the meaning of $A = X\Lambda X^{-1}$?

How is it used?

Why do we care about any of this?
Isn’t the future in ML and AI?

Unit 12

The Singular Value Decomposition

Unit Goals

- Be able to explain how matrices are used to store large data sets.
- Be able to explain how matrices can be related to images.
- Be able to explain how the singular value decomposition (SVD) relates to a sum of rank-1 matrices.
- Be able to explain how the SVD relates to the four fundamental subspaces.
- Be able to use the SVD to find a low-rank approximation to a matrix.
- Be able to use the SVD to perform principal component analysis.
- Be able to use the SVD to compute a pseudoinverse.
- Be able to use the SVD to solve linear systems that are neither full row nor full column rank.

Unit 12.1

Matrices of data.

So far … systems of equations.

- Problems involving balancing production with demand, production with available resources, economic output, and chemical equations all gave rise to systems of equations that could be expressed as:

$$A\vec{x} = \vec{b}$$

  - A represents the coefficients of the linear system—a representation of the system of equations for a model.

- Markov processes and linear dynamical systems fundamentally give rise to systems of equations of the form:

$$A\vec{x}_i = \vec{x}_{i+1}$$

  - A represents the transition coefficients of the system—again a representation of the system of equations for a model.

Data matrices.

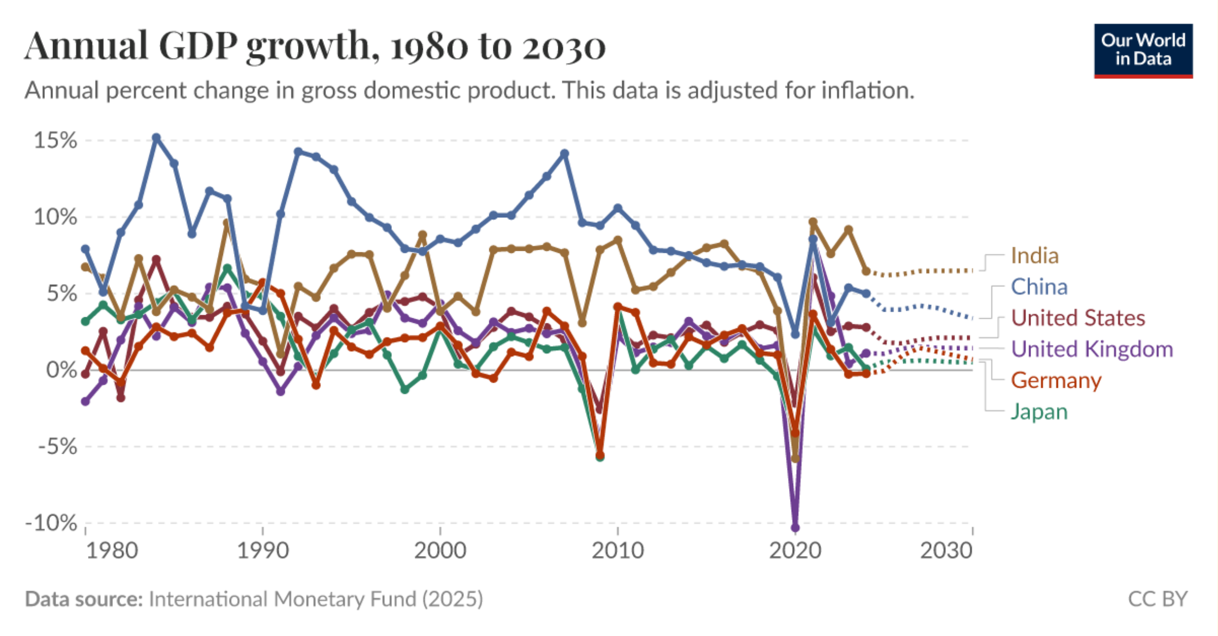
- Matrices also arise in many situations in which we tend not to think of an associated linear system or an explicit underlying mathematical model; **they are simply rectangular arrays of numbers**.
- Grading data from class (made up numbers):

	Homework	Quiz	Exam
Student 1	78%	84%	67%
Student 2	85%	76%	72%
Student 3	100%	100%	93%
Student 4	100%	40%	31%
Student 5	91%	95%	86%
Student 6	92%	87%	62%

$$\rightarrow A = \begin{bmatrix} 0.78 & 0.84 & 0.67 \\ 0.85 & 0.76 & 0.72 \\ 1.00 & 1.00 & 0.93 \\ 1.00 & 0.40 & 0.31 \\ 0.91 & 0.95 & 0.86 \\ 0.92 & 0.87 & 0.62 \end{bmatrix}$$

Big Data in Finance and Economics

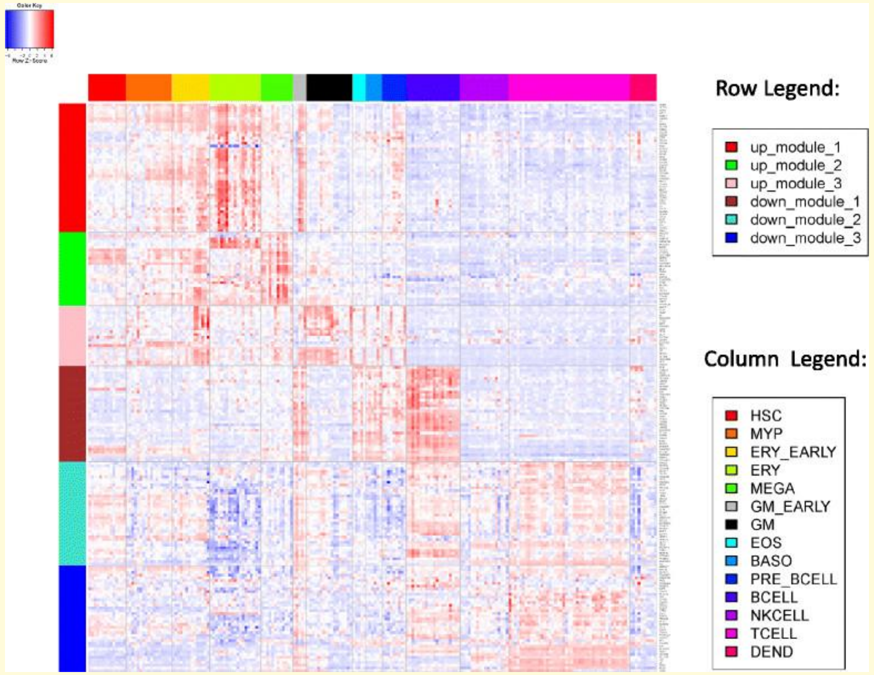
- Financial and economic time series datasets naturally lend themselves to representation in large matrices:
- Example: National GDP growth over time.



	India	China	US	…
	⋮	⋮	⋮	
2023	+9.2%	+5.4%	+2.9%	
2022	+7.6%	+3.1%	+2.5%	…
2021	+9.7%	+8.6%	+6.1%	…
2020	-5.8%	+2.3%	-2.2%	…
2019	+3.9%	+6.1%	+2.6%	…
	⋮	⋮	⋮	

Big Data in Biology

- **Many "high-throughput" experiments in biology generate big data matrices:**
- **Example: Expression levels of genes in different cell types**

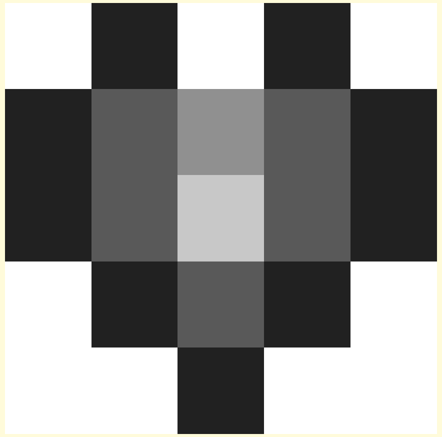


	Cell Type 1	Cell Type 2	…	Cell Type 200
Gene 1	12501	3281		2983
Gene 2	5707	6347		6532
Gene 3	102	8002		7692
Gene 4	11597	12432		11832
⋮				
Gene 30,000	5	7		11

* Data in table is completely "made-up."

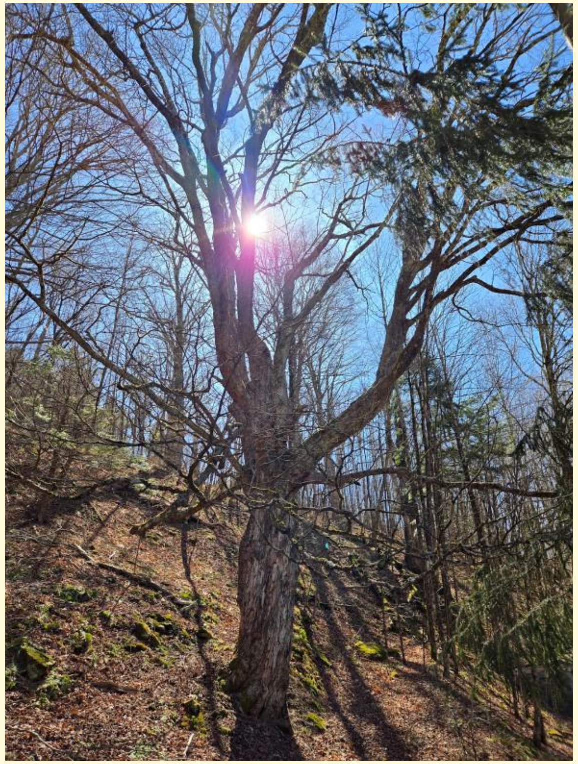
Big Data in Image Processing

- Digital images are rectangular arrays of numbers representing color and/or intensity; a pixel (picture element) is one entry in the array.
- A 12 megapixel camera records 4092x3072 arrays!



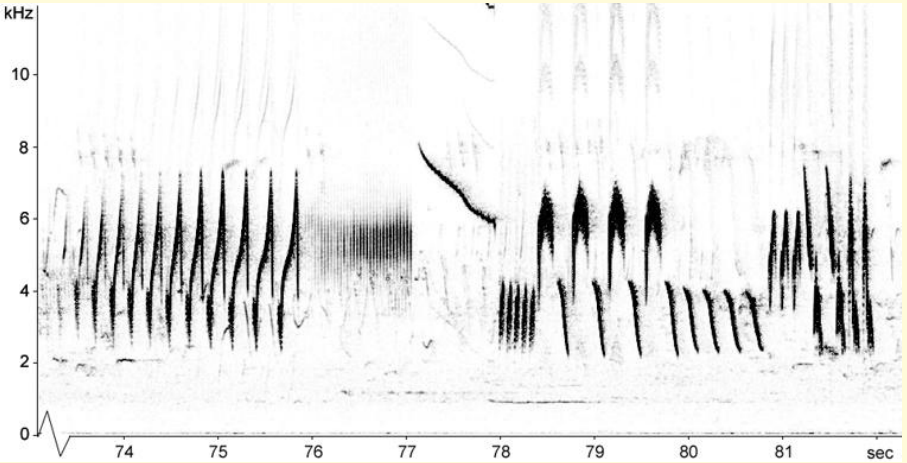
$$\rightarrow A = \begin{bmatrix} 0 & 1.0 & 0 & 1.0 & 0 \\ 1.0 & 0.7 & 0.5 & 0.7 & 1 \\ 1.0 & 0.7 & 0.3 & 0.7 & 1.0 \\ 0 & 1.0 & 0.7 & 1.0 & 0 \\ 0 & 0 & 1.0 & 0 & 0 \end{bmatrix}$$

$$\begin{matrix} A_R \\ A_G \\ A_B \end{matrix} \leftarrow$$



Big Data in Audio Processing

- **What is a sound recording?**
- **At any point in time we can measure the signal amplitude (intensity/volume) at different frequencies (tones) → a vector.**
- **This changes over time → a matrix!**



A recording of a bird call.

$$\rightarrow A = [a_{ij}],\quad \begin{array}{l} a_{ij} \leftarrow amplitude \\ i \leftarrow frequency \\ j \leftarrow time \end{array}$$

- Each column is the signal vector at a certain time.
- Each row shows the signal for one frequency (tone) over time.

Data, information and knowledge

- One of the central problems in data science is how to translate from data to knowledge and understanding.
- **A big question: how much information is actually in a data set?**
  - What are the “essential features”?
  - What are the subtle details?
  - What is random noise (error)?

These ideas are essential to building intuitive understanding, to recognizing similarities between sets of data, in efficient storage and transmission, and in the development of next generation tools.

Data, information and knowledge

- Example: The rows and columns in a data series may be highly correlated, or not; these can yield very different meanings!
- What do these sets of (fake) classroom data tell us?

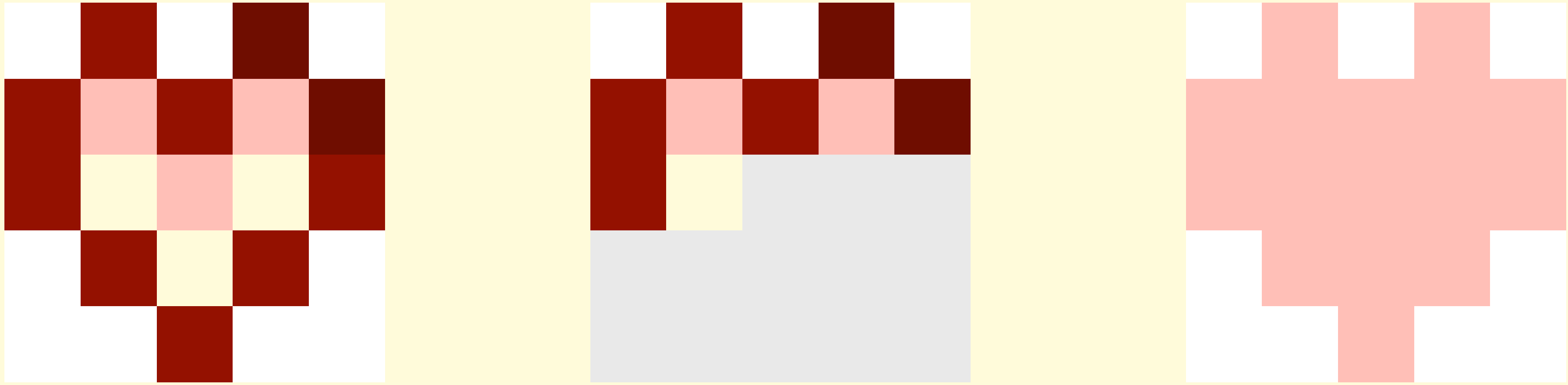
ID	HW	Quiz	Exam
1	20%	80%	50%
2	20%	50%	80%
3	80%	20%	50%
4	50%	20%	80%
5	80%	50%	20%
6	50%	80%	20%

ID	HW	Quiz	Exam
1	80%	50%	20%
2	80%	50%	20%
3	80%	50%	20%
4	80%	50%	20%
5	80%	50%	20%
6	80%	50%	20%

ID	HW	Quiz	Exam
1	20%	20%	20%
2	20%	20%	20%
3	50%	50%	50%
4	50%	50%	50%
5	80%	80%	80%
6	80%	80%	80%

Data, information and knowledge

- Example: Partial data sets are not all equally useful.
- Which image with reduced information content is more representative of the first image?
- How might this be relevant to streaming video?



Data, information and knowledge

A few goals:

- Be able to quantify correlations.
- Be able to identify “big picture” trends.
- Be able to create useful “low information content” representations.

Unit 12.2

The Singular Value Decomposition

Three (four) powerful decompositions

- **LU (Elimination, Solving consistent systems):**

$$A = LU:\ A\vec{x} = \vec{b} \rightarrow L\vec{c} = \vec{b}, U\vec{x} = \vec{c}$$

- **QR (Orthonormalization, Solving inconsistent systems):**

$$A = QR:\ A\vec{x} \approx \vec{b} \rightarrow R\vec{x} = Q^T Q\vec{b}$$

- **EVD (Eigenvalue problem, Solving dynamical systems):**

$$A = X\Lambda X^{-1}:\ A\vec{x}_i = \vec{x}_{i+1} \rightarrow \vec{x}_k = A^k\vec{x}_0 = X\Lambda^k X^{-1}\vec{x}_0$$

- **Spectral Decomposition (EVD for symmetric matrices):**

$$S = Q\Lambda Q^T:\ \vec{x}_k = S^k\vec{x}_0 = Q\Lambda^k Q^T\vec{x}_0$$

Three (four) powerful decompositions

- These decompositions are effective because the component matrices are simpler in some fundamental way:
  - Triangular: L, U, R
  - Orthogonal: Q
  - Diagonal: Λ

$$A = LU, A = QR, A = X\Lambda X^{-1}, S = Q\Lambda Q^T$$

Another decomposition:

- **LU:** $A = LU$ **or** $PA = LU$
- **QR:** $A = QR$
- **EVD:** $A = X\Lambda X^{-1}$
- **Spectral Decomposition:** $S = Q\Lambda Q^T$

Singular Value Decomposition: $\boldsymbol{A = U\Sigma V^T}$

- U **and** V **are both orthogonal matrices:** $U^T = U^{-1}, V^T = V^{-1}$
- Σ **is diagonal (and the same shape as** A**)**
- **Exists (nearly unique) for *any* matrix!**

The Singular Value Decomposition

$$A = U\Sigma V^T$$

- A matrix decomposition that is particularly useful in dealing with data matrices.
  - Can be used to find "reduced rank" approximations of matrices.
  - Can be used to identify the "principal components" of a data set.
- It is also applicable to solving linear systems of the types we have already seen:
  - Can be used to compute a pseudo-inverse for matrices that are neither full column nor full row rank.
  - Can be used to minimize the error when computing the inverse of "near singular" matrices.
- Can aid in interpreting matrices as linear mappings.

Prelude: Rank-1 matrices.

- Matrices with rank 1 are among the simplest (lowest information) matrices.
- All rows are scalar multiples of each other (as are all columns).
- Only two vectors of information – a reference row (column) and a set of scalings.
- Generated via an outer product is $\vec{u}\vec{v}^T$; *c.f.* the inner (dot) product, $\vec{u}^T\vec{v}$, a scalar.

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 \\ 2 & 0 & 4 \\ 3 & 0 & 6 \\ 4 & 0 & 8 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \end{bmatrix} \qquad \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 0 & 0 & 0 \\ 3 & 3 & 6 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 & 2 \end{bmatrix}$$

Matrices as sums of rank-1 matrices.

- Every matrix of rank r can be written as a sum of r rank-1 matrices:

Examples:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ -1 & -1 & -1 & -1 \\ -1 & -1 & -1 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 5 \\ 2 & 3 & 5 \\ 3 & 2 & 5 \\ 4 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 3 & 0 & 3 \\ 4 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 4 & 4 \\ 0 & 3 & 3 \\ 0 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

Matrices as sums of rank-1 matrices.

- **This sum is not unique:**

$$\begin{bmatrix}1&2&3\\0&1&2\\0&0&1\end{bmatrix}=\begin{bmatrix}1&2&3\\1&2&3\\1&2&3\end{bmatrix}+\begin{bmatrix}0&0&0\\-1&-1&-1\\-1&-1&-1\end{bmatrix}+\begin{bmatrix}0&0&0\\0&0&0\\0&-1&-1\end{bmatrix}$$

$$\begin{bmatrix}1&2&3\\0&1&2\\0&0&1\end{bmatrix}=\begin{bmatrix}1&2&3\\0&0&0\\0&0&0\end{bmatrix}+\begin{bmatrix}0&0&0\\0&1&2\\0&0&0\end{bmatrix}+\begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}$$

$$\begin{bmatrix}1&2&3\\0&1&2\\0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix}+\begin{bmatrix}0&2&0\\0&1&0\\0&0&0\end{bmatrix}+\begin{bmatrix}0&0&3\\0&0&2\\0&0&1\end{bmatrix}$$

Matrices as sums of outer products

- **Every matrix of rank r can be written as a sum of r rank-1 matrices:**

$$\begin{bmatrix} 1 & 4 & 5 \\ 2 & 3 & 5 \\ 3 & 2 & 5 \\ 4 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 3 & 0 & 3 \\ 4 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 4 & 4 \\ 0 & 3 & 3 \\ 0 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

- **Every rank-1 matrix can be written as an outer product:**

$$\begin{bmatrix} 1 & 4 & 5 \\ 2 & 3 & 5 \\ 3 & 2 & 5 \\ 4 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

- **Every matrix of rank r can be written as a sum of r outer products:**

$$\forall A \in \mathbb{R}^{m \times n}: rank(A) = r \rightarrow A = \sum_{i=1}^{r} \vec{x}_i \vec{y}_i^T, \vec{x}_i \in \mathbb{R}^m, \vec{y}_i \in \mathbb{R}^n$$

Not unique!

Matrices as sums of outer products

- **Every matrix of rank r can be written as a non-unique sum of r outer products:**

$$\forall A \in \mathbb{R}^{m\times n}: rank(A) = r \to A = \sum_{i=1}^{r} \vec{x}_i \vec{y}_i^T, \vec{x}_i \in \mathbb{R}^m, \vec{y}_i \in \mathbb{R}^n$$

- **Any vector can be written as a linear combination of orthonormal basis vectors. If $span(\{\vec{u}_i\}) = \mathbb{R}^m$ and $span(\{\vec{v}_i\}) = \mathbb{R}^n$:**

$$\vec{x}_i \in \mathbb{R}^m \to \vec{x}_i \in \sum_{j=1}^{m} a_i \vec{u}_i, \qquad \vec{y}_i \in \mathbb{R}^n \to \vec{y}_i \in \sum_{j=1}^{n} b_i \vec{v}_i$$

- **Claim: Every matrix of rank r can be written as a linear combination of r outer products of orthonormal vectors with positive coefficients.**

$$A = \sum_{i=1}^{r} \sigma_i \vec{u}_i \vec{v}_i^T, \sigma_i > 0$$

To prove this: Find the set of $\vec{u}_i$ and $\vec{v}_i$.

The Singular Value Decomposition

- Define two square matrices whose columns are orthonormal bases for $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively:

$$U = [\vec{u}_1 \quad \vec{u}_2 \quad \cdots \quad \vec{u}_m], V = [\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n]$$

  - Note that U is $m \times m$ and V is $n \times n$, and both are full rank (non-singular).

- Define a diagonal matrix of the same size as A ($m \times n$):

$$\Sigma = diag_A([\sigma_1, \sigma_2, \cdots, \sigma_r, 0, \cdots]), rank(\Sigma) = r \leq \min(m, n)$$

- Then $U\Sigma V^T$ gives a matrix the same size and rank as A:

$$(m \times m)(m \times n)(n \times n) \Rightarrow (m \times n)$$
$$rank\left(U\Sigma V^T\right) = rank(\Sigma) = r$$

The Singular Value Decomposition

$$U\Sigma V^T = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \cdots \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \ddots \end{bmatrix} \begin{bmatrix} \vec{v}_1^T \\ \vec{v}_2^T \\ \vdots \end{bmatrix}$$

$$U\Sigma V^T = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \cdots \end{bmatrix} \begin{bmatrix} \sigma_1 \vec{v}_1^T \\ \sigma_2 \vec{v}_2^T \\ \vdots \end{bmatrix}$$

$$U\Sigma V^T = \vec{u}_1 \sigma_1 \vec{v}_1^T + \vec{u}_2 \sigma_2 \vec{v}_2^T + \cdots$$

$$if\, \sigma_i > 0, \forall i \in [1, r], \sigma_i = 0, \forall i > r \Rightarrow U\Sigma V^T = \sum_{i=1}^{r} \sigma_i \vec{u}_i \vec{v}_i^T$$

Sum of outer products

The Singular Value Decomposition

- **Can we find orthonormal bases to define U and V such that $A = U\Sigma V^T$?**
- **The product of a matrix and its transpose is always symmetric:**

$$\left(A^T A\right)^T = A^T\left(A^T\right)^T = A^T\,A \rightarrow A^T A = S$$

- **Symmetric matrices always have a spectral decomposition:**

$$A^T A = S = Q\Lambda Q^T, \qquad Q^T Q = QQ^T = I, \lambda_i \in \mathbb{R}$$

- **The product of a matrix and its transpose is always positive semi-definite:**

$$\vec{x}^T\left(A^T A\right)\vec{x} = \left(\vec{x}^T A^T\right)A\vec{x} = (A\vec{x})^T A\vec{x} = ||A\vec{x}||^2 \geq 0, \forall\vec{x}$$

- **The eigenvalues of a positive semi-definite matrix are always non-negative:**

$$M\vec{x} = \lambda\vec{x} \rightarrow \vec{x}^T M\vec{x} = \lambda\vec{x}^T\vec{x} = \lambda||\vec{x}||^2 \geq 0 \;\rightarrow\; \lambda \geq 0$$

$$A^T A = Q\Lambda Q^T, \qquad \lambda_i \geq 0, \forall i$$

The Singular Value Decomposition

- **Can we find orthonormal bases to define U and V such that $A = U\Sigma V^T$?**
- **The product of a matrix and its transpose always has a spectral decomposition with non-negative eigenvalues:**

$$A^T A = Q\Lambda Q^T, Q^T Q = QQ^T = I, \lambda_i \geq 0, \forall i$$

Note: $rank(\Lambda) = rank\left(A^T A\right) = rank(A) = r \rightarrow \begin{cases} \lambda_i > 0, i \leq r \\ \lambda_i = 0, i > r \end{cases}$

- **Assume the SVD exists:**

$$A = U\Sigma V^T, U^T = U^{-1}, V^T = V^{-1}$$

$$A^T A = \left(U\Sigma V^T\right)^T U\Sigma V^T = \left(V\Sigma^T U^T\right) U\Sigma V^T = V\Sigma^T U^T U\Sigma V^T = V\Sigma^T \Sigma V^T$$

$$def: V \equiv Q, \qquad \Sigma^T \Sigma \equiv \Lambda, \sigma_i^2 = \lambda_i \rightarrow \sigma_i = +\sqrt{\lambda_i}$$

The Singular Value Decomposition

- **Can we find orthonormal bases to define U and V such that $A = U\Sigma V^T$?**
- **Assume the SVD exists: $A = U\Sigma V^T \rightarrow A^T A = Q\Lambda Q^T = V\Sigma^T\Sigma V^T$**

$$V = Q, \Sigma = diag_A\left(+\sqrt{\lambda_i}\right)$$

- **The singular values are the square roots of the eigenvalues of $A^T A$, $\sigma_i = +\sqrt{\lambda_i}$.**
- **The singular vectors $\{\vec{v}_i\}$ are orthonormal eigenvectors of $A^T A$.**
- **Singular vectors $\vec{v}_i$ with $\sigma_i = \sqrt{\lambda_i} = 0$ $(i > r)$ are in the null space of A:**

$$\vec{x} \in \boldsymbol{N}(A) \rightarrow A\vec{x} = \vec{0} \rightarrow A^T A\vec{x} = A^T\vec{0} = \vec{0} = 0\vec{x}$$

- **Singular vectors $\vec{v}_i$ with $\sigma_i = +\sqrt{\lambda_i} \neq 0$ $(i \leq r)$ are in the row space of A:**

$$\boldsymbol{R}(A) = \boldsymbol{N}(A)^{\perp}, \qquad A^T A\vec{x} = \lambda\vec{x} \rightarrow A^T\vec{y} = \lambda\vec{x}, \vec{y} = A\vec{x} \rightarrow \lambda\vec{x} \in \boldsymbol{C}\left(A^T\right) \rightarrow \vec{x} = \boldsymbol{R}(A)$$

The Singular Value Decomposition

$$A = U\Sigma V^T \rightarrow A^T A = Q\Lambda Q^T = V\Sigma^T\Sigma V^T \rightarrow V = Q, \Sigma = diag_A\left(+\sqrt{\lambda_i}\right)$$

- **The singular values are the square roots of the eigenvalues of $A^T A$.**
- **The singular vectors $\{\vec{v}_i\}$ are orthonormal eigenvectors of $A^T A$.**
- **Singular vectors $\vec{v}_i$ with $\sigma_i = 0$ ($i > r$) are in the null space of A; singular vectors $\vec{v}_i$ with $\sigma_i \neq 0$ ($i \leq r$) are in the row space of A.**

$$V = [V_R \quad V_N], \qquad V_R = [\vec{v}_1 \quad \cdots \quad \vec{v}_r], V_N = [\vec{v}_{r+1} \quad \cdots \quad \vec{v}_n]$$

$$\Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}, \qquad \hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}, \sigma_i > 0, \forall i \leq r$$

The Singular Value Decomposition

- **Assuming the SVD exists:**

$$A = U\Sigma V^T \rightarrow A^T A = Q\Lambda Q^T = V\Sigma^T\Sigma V^T \rightarrow V = Q, \Sigma = diag_A\left(+\sqrt{\lambda_i}\right)$$

$$V = [V_R \quad V_N], \qquad \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}$$

$$A = U\Sigma V^T \rightarrow AV = U\Sigma$$

$$AV = A[V_R \quad V_N] = [AV_R \quad AV_N] = [AV_R \quad 0]$$

$$U\Sigma = [U_{i\le r} \quad U_{i>r}]\begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix} = [U_{i\le r}\hat{\Sigma} \quad 0]$$

The Singular Value Decomposition

- **Assuming the SVD exists:**

$$A = U\Sigma V^T \rightarrow A^T A = Q\Lambda Q^T = V\Sigma^T\Sigma V^T \rightarrow V = Q, \Sigma = diag_A\left(+\sqrt{\lambda_i}\right)$$

$$V = [V_R \quad V_N], \qquad \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}$$

$$A = U\Sigma V^T \rightarrow AV = U\Sigma \rightarrow [AV_R \quad 0] = \begin{bmatrix} U_{i\le r}\hat{\Sigma} & 0 \end{bmatrix}$$

- **For $(i \le r)$, the singular vectors $\vec{u}_i$ are in the column space of A and can be found by scaling $A\vec{v}_i$:**

$$A\vec{v}_i = \sigma_i \vec{u}_i \rightarrow \vec{u}_i = \left(\frac{1}{\sigma_i}\right) A\vec{v}_i \in \boldsymbol{C}(A)$$

- **For $(i > r)$, the singular vectors $\vec{u}_i$ are in the left null space of A and can be found by standard methods:**

$$\boldsymbol{N}\left(A^T\right) = \boldsymbol{C}(A)^{\perp}, \qquad A^T = V\Sigma^T U^T \rightarrow A^T U = V\Sigma^T = \begin{bmatrix} V_R\hat{\Sigma} & 0 \end{bmatrix} = \begin{bmatrix} A^T U_{i\le r} & A^T U_{i>r} \end{bmatrix}$$

The Singular Value Decomposition

We are able to define orthonormal bases consistent with $A = U\Sigma V^T$.

$$A = U\Sigma V^T \rightarrow V = \begin{bmatrix} V_R & V_N \end{bmatrix}, U = \begin{bmatrix} U_C & U_{LN} \end{bmatrix}, \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}$$

- The $r = rank(A)$ singular values σ_i in $\hat{\Sigma}$ are the square roots of the non-zero eigenvalues of $A^T A$, $\sigma_i = +\sqrt{\lambda_i}$ (repeated for multiplicities over 1).
- The r singular vectors $\vec{v}_i$ of V_R are orthonormal eigenvectors of $A^T A$ with non-zero eigenvalues; they are a basis for $\boldsymbol{R}(A)$.
- The r singular vectors of U_C are derived from V_R by $\vec{u}_i = \left({}^1/_{\sigma_i}\right)A\vec{v}_i$; they are an orthonormal basis for $\boldsymbol{C}(A)$.
- The $n - r$ vectors of V_N are ***any*** orthonormal basis for $\boldsymbol{N}(A)$.
- The $m - r$ vectors of U_{LN} are ***any*** orthonormal basis for $\boldsymbol{N}(A^T)$.

The Singular Value Decomposition

$$A = U\Sigma V^T \to V = \begin{bmatrix} V_R & V_N \end{bmatrix}, U = \begin{bmatrix} U_C & U_{LN} \end{bmatrix}, \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}$$

- **The singular value decomposition is *almost* unique.**
- **We could arrange the singular values in $\hat{\Sigma}$ in any order, but by convention we always order them by decreasing magnitude:**

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$$

- **For non-repeated singular values there are two possible $\vec{v}_i$ with norm 1, differing by a sign; this choice determines $\vec{u}_i$.**
- **If any of the singular values σ_i are repeated, *any* orthonormal basis for the eigenspace can be used to chose $\vec{v}_i$; this choice will determine $\vec{u}_i$.**
- **If $nullity(A) > 0$, *any* orthonormal basis for $\boldsymbol{N}(A)$ can be used for V_N.**
- **If $nullity\left(A^T\right) > 0$, *any* orthonormal basis for $\boldsymbol{N}(A^T)$ can be used for U_{LN}.**

Null spaces are not needed to build A.

- **Only r terms (non-zero σ_i) in the sum of rank-1 matrices:**

$$A = \sum_{i=1}^{r} \sigma_i \vec{u}_i \vec{v}_i^T, \sigma_i > 0$$

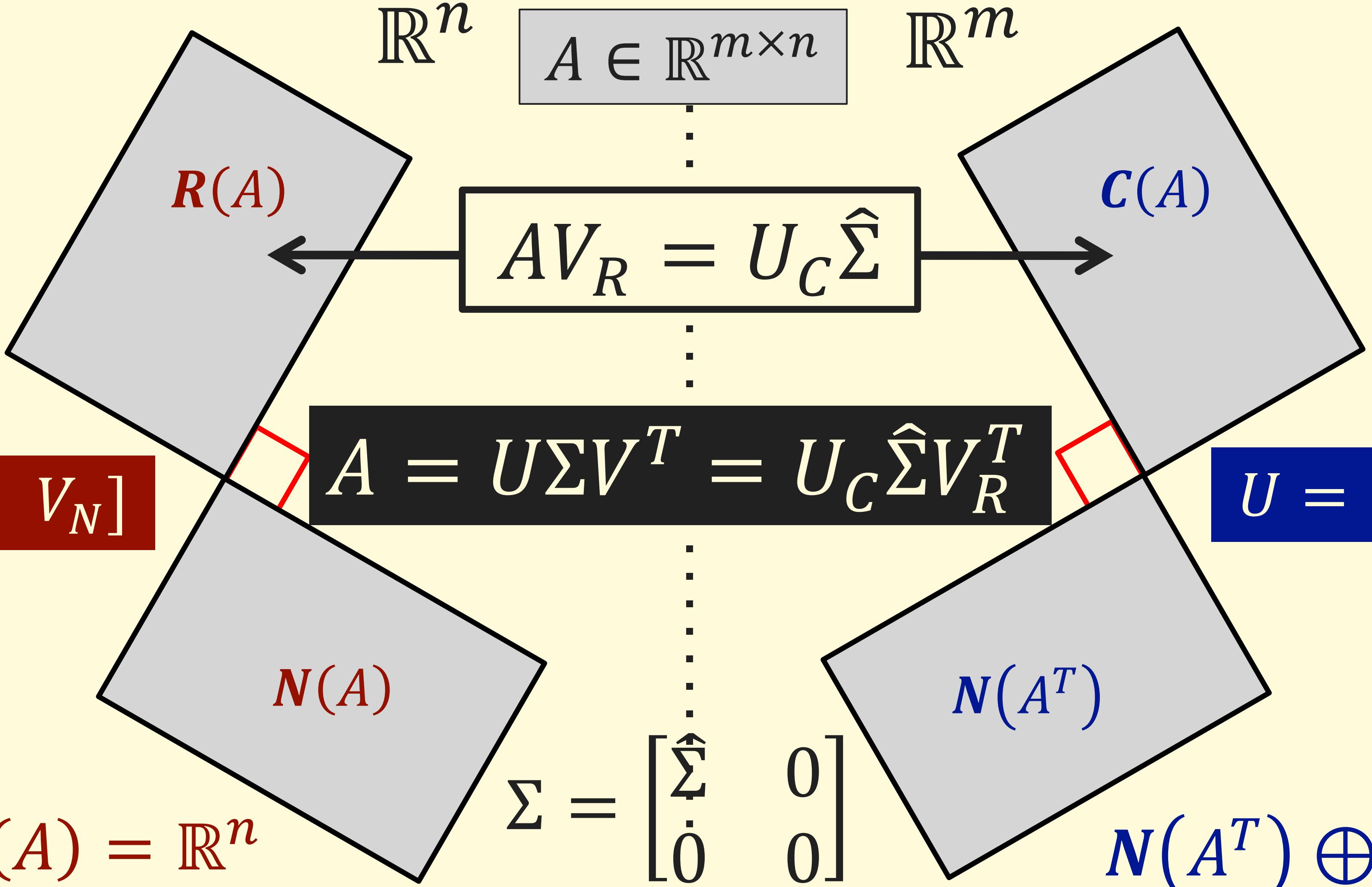
- **The full SVD was represented as:**

$$A = U\Sigma V^T, V = [V_R \quad V_N], U = [U_C \quad U_{LN}], \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

- **Multiplying out the matrices, we find that both null spaces can be ignored; only the row and column space vectors are needed to perfectly reconstruct A:**

$$A = [U_C \quad U_{LN}] \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V_R^T \\ V_N^T \end{bmatrix} = [U_C \quad U_{LN}] \begin{bmatrix} \hat{\Sigma} V_R^T \\ 0 \end{bmatrix} = U_C \hat{\Sigma} V_R^T$$

The SVD and the big picture.



$$A \in \mathbb{R}^{m\times n}$$

$$\boxed{AV_R = U_C\hat{\Sigma}}$$

$$A = U\Sigma V^T = U_C\hat{\Sigma}V_R^T$$

$$V = [V_R \quad V_N]$$

$$U = [U_C \quad U_{LN}]$$

$$\Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}$$

$$\boldsymbol{R}(A) \oplus \boldsymbol{N}(A) = \mathbb{R}^n$$

$$\boldsymbol{N}(A^T) \oplus \boldsymbol{C}(A) = \mathbb{R}^m$$

Unit 12.3.1

Worked Example Finding the SVD

Find the SVD

- **Find the singular value decomposition of the matrix:**

$$A = \begin{bmatrix} 1 & -3 & 0 \\ 3 & -1 & 0 \end{bmatrix}$$

Find the SVD

- **First, find the eigenvalues of A^TA:**

$$A = \begin{bmatrix} 1 & -3 & 0 \\ 3 & -1 & 0 \end{bmatrix} \rightarrow A^TA = \begin{bmatrix} 1 & 3 \\ -3 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 \\ 3 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 10 & -6 & 0 \\ -6 & 10 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|\lambda I - A^TA| = \begin{vmatrix} \lambda - 10 & 6 & 0 \\ 6 & \lambda - 10 & 0 \\ 0 & 0 & \lambda \end{vmatrix} = \lambda \begin{vmatrix} \lambda - 10 & 6 \\ 6 & \lambda - 10 \end{vmatrix}$$

$$= \lambda\left((\lambda - 10)^2 - 6^2\right) = \lambda\left(\lambda^2 - 20\lambda + 100 - 36\right) = \lambda(\lambda - 16)(\lambda - 4)$$

$$|\lambda I - A^TA| = 0 \rightarrow \lambda(\lambda - 16)(\lambda - 4) = 0 \rightarrow \lambda = 16,4,0$$

$$\sigma_i^2 = \lambda_i \rightarrow \sigma = 4,2 \rightarrow \Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$$

Find the SVD

$$A^T A = \begin{bmatrix} 10 & -6 & 0 \\ -6 & 10 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \lambda = 16,4,0$$

- **Then, find the eigenvectors of $A^T A$:**

$$\lambda = 16: \lambda I - A^T A = \begin{bmatrix} 6 & 6 & 0 \\ 6 & 6 & 0 \\ 0 & 0 & 16 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \to \vec{x} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \to \begin{bmatrix} -{}^{1}/_{\sqrt{2}} \\ {}^{1}/_{\sqrt{2}} \\ 0 \end{bmatrix}$$

$$\lambda = 4: \lambda I - A^T A = \begin{bmatrix} -6 & 6 & 0 \\ 6 & -6 & 0 \\ 0 & 0 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \to \vec{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \to \begin{bmatrix} {}^{1}/_{\sqrt{2}} \\ {}^{1}/_{\sqrt{2}} \\ 0 \end{bmatrix}$$

$$\lambda = 0: \lambda I - A^T A = \begin{bmatrix} -10 & 6 & 0 \\ 6 & -10 & 0 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \to \vec{x} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Find the SVD

- Then, find the eigenvectors of $A^T A$:

$$\lambda_1 = 16 \rightarrow \sigma_1 = 4, \vec{v}_1 = \begin{bmatrix} -{}^{1}\!/_{\sqrt{2}} \\ {}^{1}\!/_{\sqrt{2}} \\ 0 \end{bmatrix}, \lambda_2 = 4 \rightarrow \sigma_2 = 2, \vec{v}_2 = \begin{bmatrix} {}^{1}\!/_{\sqrt{2}} \\ {}^{1}\!/_{\sqrt{2}} \\ 0 \end{bmatrix}, \lambda_3 = 0 \rightarrow \vec{v}_n = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$V = \begin{bmatrix} -{}^{1}\!/_{\sqrt{2}} & {}^{1}\!/_{\sqrt{2}} & 0 \\ {}^{1}\!/_{\sqrt{2}} & {}^{1}\!/_{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Find the SVD

- **Finally, compute** AV:

$$AV = \begin{bmatrix} 1 & -3 & 0 \\ 3 & -1 & 0 \end{bmatrix} \begin{bmatrix} -{}^{1}/_{\sqrt{2}} & {}^{1}/_{\sqrt{2}} & 0 \\ {}^{1}/_{\sqrt{2}} & {}^{1}/_{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -{}^{4}/_{\sqrt{2}} & {}^{-2}/_{\sqrt{2}} & 0 \\ {}^{-4}/_{\sqrt{2}} & {}^{2}/_{\sqrt{2}} & 0 \end{bmatrix}$$

$$AV = U\Sigma \to \begin{bmatrix} -{}^{4}/_{\sqrt{2}} & {}^{-2}/_{\sqrt{2}} & 0 \\ {}^{-4}/_{\sqrt{2}} & {}^{2}/_{\sqrt{2}} & 0 \end{bmatrix} = U \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \to U = \begin{bmatrix} -{}^{1}/_{\sqrt{2}} & {}^{-1}/_{\sqrt{2}} \\ {}^{-1}/_{\sqrt{2}} & {}^{1}/_{\sqrt{2}} \end{bmatrix}$$

Find the SVD

- **Find the singular value decomposition of the matrix:**

$$A = \begin{bmatrix} 1 & -3 & 0 \\ 3 & -1 & 0 \end{bmatrix}$$

$$A = U\Sigma V^T$$

$$U = \begin{bmatrix} -{}^{1}/_{\sqrt{2}} & -{}^{1}/_{\sqrt{2}} \\ -{}^{1}/_{\sqrt{2}} & {}^{1}/_{\sqrt{2}} \end{bmatrix}, \Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}, V = \begin{bmatrix} -{}^{1}/_{\sqrt{2}} & {}^{1}/_{\sqrt{2}} & 0 \\ {}^{1}/_{\sqrt{2}} & {}^{1}/_{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Unit 12.3

Applications of the SVD

Reduced-rank approximations.

- Most data matrices are either full row rank or full column rank; for matrices with both many rows and many columns this rank is large:

$$A \in \mathbb{R}^{m \times n}, rank(A) = \min(m, n)$$

- However, many data matrices can be well-approximated by a matrix of significantly lower rank:

$$\hat{A}_k \in \mathbb{R}^{m \times n}, rank\left(\hat{A}_k\right) = k \ll \min(m, n), \hat{A}_k \approx A$$

- This low-rank approximation may be thought of a "big picture" view of the matrix, capturing the essential features, but missing subtle details and/or noise.

Reduced-rank approximations.

- We begin with the view of as a sum of rank-1 matrices, expressed in terms of the singular values/vectors:

$$A = \sum_{i=1}^{r} \sigma_i \vec{u}_i \vec{v}_i^T \, , \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$$

- $\{\vec{u}_i\}$ and $\{\vec{v}_i\}$ are both orthonormal sets, so all $\vec{u}_i \vec{v}_i^T$ will be of similar magnitudes; the contribution of each term to the overall sum will be dominated by σ_i. We just truncate the sum:

$$\hat{A}_k \equiv \sum_{i=1}^{k} \sigma_i \vec{u}_i \vec{v}_i^T$$

Reduced-rank approximations.

- In summation notation we have: $\hat{A}_k \equiv \sum_{i=1}^{k} \sigma_i \vec{u}_i \vec{v}_i^T$
- The full SVD was represented as:

$$A = U\Sigma V^T, V = \begin{bmatrix} V_R & V_N \end{bmatrix}, U = \begin{bmatrix} U_C & U_{LN} \end{bmatrix}, \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

- We can define a reduced-rank SVD simply by changing Σ; we zero out all the singular values after the first k:

$$\hat{A}_k = U\Sigma_k V^T, \Sigma_k = \begin{bmatrix} \hat{\Sigma}_k & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma}_k = \begin{bmatrix} \sigma_1' & & \\ & \ddots & \\ & & \sigma_r' \end{bmatrix}, \sigma_i' = \sigma_i, \forall i \leq k, \sigma_i' = 0, \forall i > k$$

Reduced-rank approximations.

- In a simplified notation we can write:

$$\hat{A}_k = U\Sigma_k V^T$$

$$\Sigma_k = \begin{bmatrix} \hat{\Sigma}_k & 0 \\ 0 & 0 \end{bmatrix}, \qquad \hat{\Sigma}_k = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_k \end{bmatrix}$$

- $\hat{A}_k$ is the **best** rank-k approximation of A, in a "least-squares" sense:

$$\hat{A}_k = \min_{X, rank(X)=k} \left(\sum_{i,j} \left(A_{ij} - X_{ij}\right)^2 \right)$$

Efficient storage of A and $\hat{A}_k$

- The full SVD was represented as:

$$A = U\Sigma V^T, V = \begin{bmatrix} V_R & V_N \end{bmatrix}, U = \begin{bmatrix} U_C & U_{LN} \end{bmatrix}, \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

- Multiplying out the matrices, we find that both null spaces can be ignored; only the row and column space vectors are needed to perfectly reconstruct A:

$$A = \begin{bmatrix} U_C & U_{LN} \end{bmatrix} \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V_R^T \\ V_N^T \end{bmatrix} = \begin{bmatrix} U_C & U_{LN} \end{bmatrix} \begin{bmatrix} \hat{\Sigma} V_R^T \\ 0 \end{bmatrix} = U_C \hat{\Sigma} V_R^T$$

- A rank-r matrix can be stored as r vectors of length m, r vectors of length n and r singular values: total storage is $mr + nr + r = r(m + n + 1)$

$$r \ll \min(m, n) \rightarrow r(m + n + 1) \ll mn$$
$$r = \min(m, n) \rightarrow r(m + n + 1) > mn$$

Efficient storage of A and $\hat{A}_k$

- The reduced-rank SVD can be represented as:

$$\hat{A}_k = U\Sigma_k V^T, V = \begin{bmatrix}\hat{V}_k & \hat{V}_N\end{bmatrix}, U = \begin{bmatrix}\hat{U}_k & \hat{U}_{LN}\end{bmatrix}, \Sigma_k = \begin{bmatrix}\hat{\Sigma}_k & 0 \\ 0 & 0\end{bmatrix}, \hat{\Sigma}_k = \begin{bmatrix}\sigma_1 & & \\ & \ddots & \\ & & \sigma_k\end{bmatrix}$$

- Here $\hat{V}_k$ and $\hat{U}_k$ contain only the first k columns of V_R and U_C; the others have been added to V_N and U_{LN} to form $\hat{V}_N$ and $\hat{U}_{LN}$. Then:

$$\hat{A}_k = \begin{bmatrix}\hat{U}_k & \hat{U}_{LN}\end{bmatrix}\begin{bmatrix}\hat{\Sigma}_k & 0 \\ 0 & 0\end{bmatrix}\begin{bmatrix}\hat{V}_k^T \\ \hat{V}_N^T\end{bmatrix} = \begin{bmatrix}\hat{U}_k & \hat{U}_{LN}\end{bmatrix}\begin{bmatrix}\hat{\Sigma}_k\hat{V}_k^T \\ \hat{V}_N^T\end{bmatrix} = \hat{U}_k\hat{\Sigma}_k\hat{V}_k^T$$

- $\hat{A}_k$ can be stored as k vectors of length m, k vectors of length n and k singular values: total storage is $mk + nk + k = k(m+n+1)$:

$$k \ll \min(m,n) \rightarrow k(m+n+1) \ll mn$$

Comparing A and $\hat{A}_k$ with image data.

- Demonstration website:
- https://timbaumann.info/svd-image-compression-demo/
- At what "k" can you correctly guess the image?

Principal component analysis.

- Consider a data matrix with each column corresponding to individual "samples" and each row corresponding to a different measurable variable:

$$A_o = \begin{bmatrix} x_1 & x_2 & x_3 & & x_n \\ y_1 & y_2 & y_3 & \cdots & y_n \\ z_1 & z_2 & z_3 & & z_n \end{bmatrix}$$

- A common statistical analysis involves computing (co-)variances:

$$var(x) = \left(\frac{1}{n-1}\right)\sum_{i=1}^{n}(x_i - \langle x \rangle)^2$$

Describes how much "spread" there is in "x".

$$cov(x,y) = \left(\frac{1}{n-1}\right)\sum_{i=1}^{n}(x_i - \langle x \rangle)(y_i - \langle y \rangle)$$

Describes how much "x" and "y" vary in a correlated manner.

Principal component analysis.

- If we "center" the data by subtracting the mean from each row:

$$A = \begin{bmatrix} x_1 - \langle x \rangle & x_2 - \langle x \rangle & x_3 - \langle x \rangle & & x_n - \langle x \rangle \\ y_1 - \langle y \rangle & y_2 - \langle y \rangle & y_3 - \langle y \rangle & \cdots & y_n - \langle y \rangle \\ z_1 - \langle z \rangle & z_2 - \langle z \rangle & z_3 - \langle z \rangle & & z_n - \langle z \rangle \end{bmatrix}$$

- We can compute all the variances and covariances with a matrix expression:

$$Cov(A) = S = \left(\frac{1}{n-1}\right) AA^T$$

Covariance matrix!

Principal component analysis.

- For a (row) mean-shifted data matrix, the covariance matrix is:

$$Cov(A) = S = \left(\frac{1}{n-1}\right)AA^T$$

- The total variance in the data is the sum of the eigenvalues of the covariance matrix:

$$T = \lambda_1 + \lambda_2 + \cdots + \lambda_n = \lambda_1 + \lambda_2 + \cdots + \lambda_r, rank(A) = r$$

- The covariance matrix involves the same product used to derive singular values of A:

$$\lambda_i = \frac{\sigma_i^2}{n-1} \rightarrow T = \left(\frac{1}{n-1}\right)\left(\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2\right)$$

Principal component analysis.

- For a (row) mean-shifted data matrix, total variance in the data is given by the sum of the singular values (scaled by the sample size):

$$T = \left(\frac{1}{n-1}\right)\left(\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2\right)$$

- The largest singular values explain most of the variance in the data.

$$T \approx T_k = \left(\frac{1}{n-1}\right)\left(\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_k^2\right), k < r$$

- Just like a low-rank approximation: $\hat{A}_k = \hat{U}_k \hat{\Sigma}_k \hat{V}_k^T$
- The singular vectors $\vec{u}_i$ tell us the directions in which the majority of the variance occurs, these are the so-called **principal components**.

Principal component analysis.

- The singular vectors $\vec{u}_i$ tell us the directions in which the majority of the variance occurs, these are the so-called **principal components**.
- To find a best fit line, we follow the single direction of greatest variance: the first principal component.
- To find a best fit plane, we consider the two directions of greatest variance: the first and second principal components.
- The first "k" principal components is the k-dimensional subspace which best (in a least-squares sense) satisfies the data.
- Note: These lines, planes, etc. fit the mean-shifted data and pass through the origin; simply translating them by the means gives the best-fit to the original data.

Pseudoinverses and the SVD

- **The full SVD was represented as:**

$$A = U\Sigma V^T, V = \begin{bmatrix} V_R & V_N \end{bmatrix}, U = \begin{bmatrix} U_C & U_{LN} \end{bmatrix}, \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

- **If A is square and nonsingular:**

$$A^{-1} = \left(U\Sigma V^T\right)^{-1} = \left(V^T\right)^{-1}\Sigma^{-1}U^{-1} = V\Sigma^{-1}U^T$$

$$\Sigma = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{bmatrix} \rightarrow \Sigma^{-1} = \begin{bmatrix} 1/\sigma_1 & & \\ & \ddots & \\ & & 1/\sigma_n \end{bmatrix}$$

Pseudoinverses and the SVD

- **The full SVD was represented as:**

$$A = U\Sigma V^T, V = [V_R \quad V_N], U = [U_C \quad U_{LN}], \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

- **If A is either non-square or singular, define:**

$$\Sigma^+ \equiv \begin{bmatrix} \hat{\Sigma}^{-1} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma}^{-1} = \begin{bmatrix} 1/\sigma_1 & & \\ & \ddots & \\ & & 1/\sigma_r \end{bmatrix}$$

$$A^+ \equiv V\Sigma^+U^T = V_R\hat{\Sigma}^{-1}U_C^T$$

Pseudoinverses and the SVD

- The product AA^+ **almost** gives the identity:

$$AA^+ = U\Sigma V^T V\Sigma^+ U^T = U\Sigma\Sigma^+ U^T$$

$$\Sigma\Sigma^+ = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{\Sigma}^{-1} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}, UU^T = I_m$$

$$AA^+ = \begin{bmatrix} U_C & U_{LN} \end{bmatrix} \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} U_C^T \\ U_{LN}^T \end{bmatrix} = U_C U_C^T$$

- $P_C = U_C U_C^T = AA^+$ is the projection matrix onto the column space.
- This will be exactly the $(m \times m)$ identity matrix if (and only if) A is full row rank (right inverse).

Pseudoinverses and the SVD

- The product A^+A also **almost** gives the identity:

$$A^+A = V\Sigma^+U^TU\Sigma V^T = V\Sigma^+\Sigma V^T$$

$$\Sigma^+\Sigma = \begin{bmatrix} \hat{\Sigma}^{-1} & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}, VV^T = I_n$$

$$A^+A = \begin{bmatrix} V_R & V_N \end{bmatrix}\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} V_R^T \\ V_N^T \end{bmatrix} = V_RV_R^T$$

- $P_R = V_RV_R^T = A^+A$ is the projection matrix onto the row space.
- This will be exactly the $(n \times n)$ identity matrix if (and only if) A is full column rank (left inverse).

Pseudoinverses and the SVD

- $P_C = U_C U_C^T = AA^+$ is the projection matrix onto the column space.
- The best least-squares solution to an inconsistent system is found by solving:

$$A\vec{x} = proj_{\boldsymbol{C}(A)}\vec{b} = P_C\vec{b} \rightarrow A\vec{x} = AA^+\vec{b}, A^+ \equiv V\Sigma^+U^T$$

$$A\vec{x} = AA^+\vec{b} \rightarrow U_C\hat{\Sigma}V_R^T\vec{x} = U_CU_C^T\vec{b} \rightarrow V_R^T\vec{x} = \hat{\Sigma}^{-1}U_C^T\vec{b}$$

$$V_RV_R^T\vec{x} = V_R\hat{\Sigma}^{-1}U_C^T\vec{b} \rightarrow P_R\vec{x} = V_R\hat{\Sigma}^{-1}U_C^T\vec{b}$$

$$\vec{x}_r = P_R\vec{x} = A^+\vec{b}$$

- $\vec{x}_r$ is the projection of the solution(s) onto the row space of A.
- This is the solution to $A\vec{x} = proj_{\boldsymbol{C}(A)}\vec{b}$ that is in the row space of A.

Pseudoinverses and the SVD

- The pseudoinverse gives the solution to $A\vec{x} = proj_{C(A)}\vec{b}$ that is in the row space of A.

$$\vec{x}_r = A^+\vec{b}, A^+ \equiv V\Sigma^+U^T$$

- The general solution is $\vec{x} = \vec{x}_r + \vec{x}_n, \forall \vec{x}_n \in \boldsymbol{N}(A)$.
- The row space and null space are orthogonal: $\vec{x}_r \perp \vec{x}_n$.
- $\vec{x}_r$ is the shortest of any solution: $\|\vec{x}\|^2 = \|\vec{x}_r\|^2 + \|\vec{x}_n\|^2$.
- Given $A\vec{x} = \vec{b}$ the pseudoinverse gives the shortest possible solution to the system $A\vec{x} = \vec{p}$ that has a right-hand side $\vec{p}$ that is the shortest possible distance from $\vec{b}$.

Unit 12.3.1

Worked Examples using the SVD

Find a low-rank approximation

- Find the best (in a least-squares sense) rank-2 approximation to the following matrix with a given SVD:

$$A = \begin{bmatrix} 3 & -2.25 & -1.75 & 3 \\ 3 & -1.75 & -2.25 & 3 \\ 2.05 & -3 & -3 & 1.95 \\ 1.95 & -3 & -3 & 2.05 \end{bmatrix}$$

$$A = \begin{bmatrix} 1/2 & 1/2 & -1/\sqrt{2} & 0 \\ 1/2 & 1/2 & 1/\sqrt{2} & 0 \\ 1/2 & -1/2 & 0 & -1/\sqrt{2} \\ 1/2 & -1/2 & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 10 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1/10 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 & -1/2 & 1/2 \\ 1/2 & 1/2 & 1/2 & 1/2 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 0 & 0 & 1/\sqrt{2} \end{bmatrix}$$

Find a low-rank approximation

- Zero out all but the two largest singular values

$$\hat{A}_2 = \begin{bmatrix} 1/2 & 1/2 & -1/\sqrt{2} & 0 \\ 1/2 & 1/2 & 1/\sqrt{2} & 0 \\ 1/2 & -1/2 & 0 & -1/\sqrt{2} \\ 1/2 & -1/2 & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 10 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 & -1/2 & 1/2 \\ 1/2 & 1/2 & 1/2 & 1/2 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 0 & 0 & 1/\sqrt{2} \end{bmatrix}$$

Find a low-rank approximation

- Zero out all but the two largest singular values

$$\hat{A}_2 = \begin{bmatrix} 1/2 & 1/2 & -1/\sqrt{2} & 0 \\ 1/2 & 1/2 & 1/\sqrt{2} & 0 \\ 1/2 & -1/2 & 0 & -1/\sqrt{2} \\ 1/2 & -1/2 & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 5 & -5 & -5 & 5 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & -2 & -2 & 3 \\ 3 & -2 & -2 & 3 \\ 2 & -3 & -3 & 2 \\ 2 & -3 & -3 & 2 \end{bmatrix}$$

Find a low-rank approximation

- Find the best (in a least-squares sense) rank-2 approximation to the following matrix with a given SVD:

$$\hat{A}_2 = \begin{bmatrix} 3 & -2 & -2 & 3 \\ 3 & -2 & -2 & 3 \\ 2 & -3 & -3 & 2 \\ 2 & -3 & -3 & 2 \end{bmatrix}, \qquad A = \begin{bmatrix} 3 & -2.25 & -1.75 & 3 \\ 3 & -1.75 & -2.25 & 3 \\ 2.05 & -3 & -3 & 1.95 \\ 1.95 & -3 & -3 & 2.05 \end{bmatrix}$$

$$\Delta = (A - \hat{A}_2) = \begin{bmatrix} 0 & -0.25 & +0.25 & 3 \\ 0 & +0.25 & -0.25 & 3 \\ +0.05 & 0 & 0 & -0.05 \\ -0.05 & 0 & 0 & +0.05 \end{bmatrix}$$

Principal component analysis

- **Find the best-fit line and plane through origin, given the eigenvalue decomposition of AA^T**

$$A = \begin{bmatrix} -5 & -3 & -1 & 1 & 3 & 5 \\ 9 & 6 & 3 & -3 & -6 & -9 \\ -2 & -1 & 0 & 0 & 1 & 2 \end{bmatrix}$$

$$AA^T = U\Sigma^2 U^T, \qquad A = U\Sigma V^T$$

$$AA^T = \begin{bmatrix} 70 & -132 & 0 \\ -132 & 252 & 0 \\ 0 & 0 & 10 \end{bmatrix}, U = \begin{bmatrix} -0.465 & 0 & -0.465 \\ 0.885 & 0 & 0.885 \\ 0 & 1.0 & 0.67 \end{bmatrix}, \Sigma^2 = \begin{bmatrix} 321.3 & 0 & 0 \\ 0 & 10.0 & 0 \\ 0 & 0 & 0.67 \end{bmatrix}$$

Principal component analysis

- The best-fit line is defined by the largest principal component:

$$A = \begin{bmatrix} -5 & -3 & -1 & 1 & 3 & 5 \\ 9 & 6 & 3 & -3 & -6 & -9 \\ -2 & -1 & 0 & 0 & 1 & 2 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 70 & -132 & 0 \\ -132 & 252 & 0 \\ 0 & 0 & 10 \end{bmatrix}, U = \begin{bmatrix} -0.465 & 0 & -0.465 \\ 0.885 & 0 & 0.885 \\ 0 & 1.0 & 0.67 \end{bmatrix}, \Sigma^2 = \begin{bmatrix} 321.3 & 0 & 0 \\ 0 & 10.0 & 0 \\ 0 & 0 & 0.67 \end{bmatrix}$$

$$\vec{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = k \begin{bmatrix} -0.465 \\ 0.885 \\ 0 \end{bmatrix}$$

Principal component analysis

- **The best-fit plane is defined by the two largest principal components:**

$$A = \begin{bmatrix} -5 & -3 & -1 & 1 & 3 & 5 \\ 9 & 6 & 3 & -3 & -6 & -9 \\ -2 & -1 & 0 & 0 & 1 & 2 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 70 & -132 & 0 \\ -132 & 252 & 0 \\ 0 & 0 & 10 \end{bmatrix}, U = \begin{bmatrix} -0.465 & 0 & -0.465 \\ 0.885 & 0 & 0.885 \\ 0 & 1.0 & 0.67 \end{bmatrix}, \Sigma^2 = \begin{bmatrix} 321.3 & 0 & 0 \\ 0 & 10.0 & 0 \\ 0 & 0 & 0.67 \end{bmatrix}$$

$$\vec{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = k \begin{bmatrix} -0.465 \\ 0.885 \\ 0 \end{bmatrix} + l \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Solve with a pseudoinverse

- Find the general least-squares solution to the following system of equations, given the SVD of the coefficient matrix:

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}$$

$$A = U\Sigma V^T, U = \begin{bmatrix} -1/2 & 1/2 & 1/\sqrt{2} & 0 \\ -1/2 & 1/2 & -1/\sqrt{2} & 0 \\ -1/2 & -1/2 & 0 & -1/\sqrt{2} \\ -1/2 & -1/2 & 0 & 1/\sqrt{2} \end{bmatrix}, \Sigma = \begin{bmatrix} 4\sqrt{3} & 0 & 0 \\ 0 & 2\sqrt{2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, V = \begin{bmatrix} -1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ -1/\sqrt{3} & 0 & -2/\sqrt{6} \\ -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}$$

Solve with a pseudoinverse

- **The least-squares solution from the row space is given by the pseudoinverse:**

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} \rightarrow \vec{x}_r = A^+\vec{b}, A^+ = V\Sigma^+U^T$$

$$A = U\Sigma V^T, U = \begin{bmatrix} -1/2 & 1/2 & 1/\sqrt{2} & 0 \\ -1/2 & 1/2 & -1/\sqrt{2} & 0 \\ -1/2 & -1/2 & 0 & -1/\sqrt{2} \\ -1/2 & -1/2 & 0 & 1/\sqrt{2} \end{bmatrix}, \Sigma = \begin{bmatrix} 4\sqrt{3} & 0 & 0 \\ 0 & 2\sqrt{2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, V = \begin{bmatrix} -1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ -1/\sqrt{3} & 0 & -2/\sqrt{6} \\ -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}$$

Solve with a pseudoinverse

- The least-squares solution from the row space is given by the pseudoinverse:

$$A^{+} = V\Sigma^{+}U^{T} = \begin{bmatrix} -1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ -1/\sqrt{3} & 0 & -2/\sqrt{6} \\ -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix} \begin{bmatrix} 1/4\sqrt{3} & 0 & 0 & 0 \\ 0 & 1/2\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1/2 & -1/2 & -1/2 & -1/2 \\ 1/2 & 1/2 & -1/2 & -1/2 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 0 & 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} -1/12 & -1/4 & 0 & 0 \\ -1/12 & 0 & 0 & 0 \\ -1/12 & 1/4 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1/2 & -1/2 & -1/2 & -1/2 \\ 1/2 & 1/2 & -1/2 & -1/2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -1/12 & -1/12 & 1/6 & 1/6 \\ 1/24 & 1/24 & 1/24 & 1/24 \\ 1/6 & 1/6 & -1/12 & -1/12 \end{bmatrix}$$

Solve with a pseudoinverse

- **The least-squares solution from the row space is given by the pseudoinverse:**

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} \rightarrow \vec{x}_r = A^+\vec{b}, A^+ = \begin{bmatrix} -1/12 & -1/12 & 1/6 & 1/6 \\ 1/24 & 1/24 & 1/24 & 1/24 \\ 1/6 & 1/6 & -1/12 & -1/12 \end{bmatrix}$$

$$\vec{x}_r = A^+\vec{b}, A^+ = \begin{bmatrix} -1/12 & -1/12 & 1/6 & 1/6 \\ 1/24 & 1/24 & 1/24 & 1/24 \\ 1/6 & 1/6 & -1/12 & -1/12 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1/12 \\ 5/12 \\ 11/12 \end{bmatrix}$$

Solve with a pseudoinverse

- **The general least-squares solution is given by adding the null space.**

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} \rightarrow \vec{x} = \vec{x}_r + s\vec{x}_n, \forall s, \vec{x}_r = A^{+}\vec{b}, A\vec{x}_n = \vec{0}$$

$$A = U\Sigma V^T, V = \begin{bmatrix} {}^{-1}\!/_{\sqrt{3}} & {}^{-1}\!/_{\sqrt{2}} & {}^{1}\!/_{\sqrt{6}} \\ {}^{-1}\!/_{\sqrt{3}} & 0 & {}^{-2}\!/_{\sqrt{6}} \\ {}^{-1}\!/_{\sqrt{3}} & {}^{1}\!/_{\sqrt{2}} & {}^{1}\!/_{\sqrt{6}} \end{bmatrix} \rightarrow \vec{x}_n = \begin{bmatrix} {}^{1}\!/_{\sqrt{6}} \\ {}^{-2}\!/_{\sqrt{6}} \\ {}^{1}\!/_{\sqrt{6}} \end{bmatrix}$$

Solve with a pseudoinverse

- The least-squares solution from the row space is given by the pseudoinverse:

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} \rightarrow \vec{x}_r = \begin{bmatrix} -1/12 \\ 5/12 \\ 11/12 \end{bmatrix}, \vec{x}_n = \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}$$

$$\vec{x} = \vec{x}_r + s\vec{x}_n = \begin{bmatrix} -1/12 \\ 5/12 \\ 11/12 \end{bmatrix} + s \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \forall s \in \mathbb{R}$$